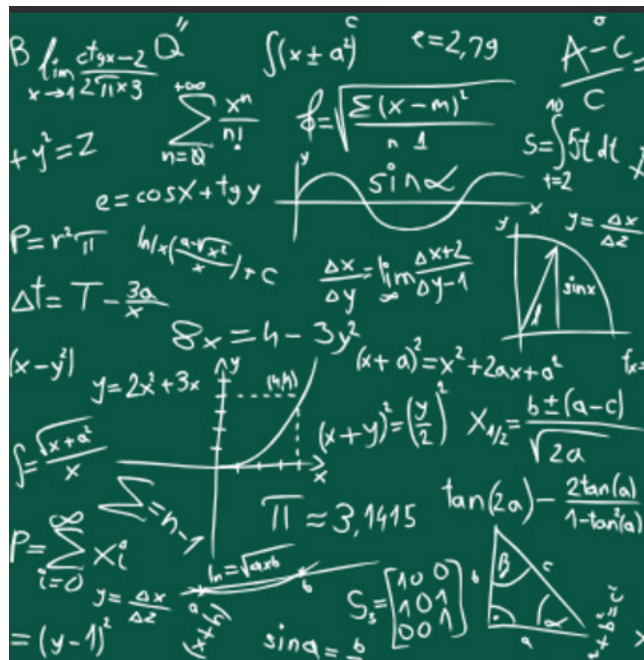


Inverse functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

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1 Invertible functions and inverse functions

- Given a real function of a real variable $f : D \rightarrow \mathbb{R}$, for every input x in the domain D there exists a unique output $y = f(x)$ in the image of the domain $f(D)$.
- If, in addition, for every output $y = f(x) \in f(D)$ there exists a unique input $x \in D$, then f is called **invertible**, and it establishes a **one-to-one correspondence** between the domain D of f and the image of the domain $f(D)$.

Definition 1: of invertible function

A function $f : D \rightarrow \mathbb{R}$ is **invertible** on the domain D if one of the following equivalent conditions holds:

$$\forall x_1, x_2 \in D, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

$$\forall x_1, x_2 \in D, \quad f(x_1) = f(x_2) \implies x_1 = x_2$$

$$\forall y \in f(D), \quad \exists! x \in D : y = f(x)$$

that is, if f is **injective**.

WARNING: the definition of invertible function given here for real functions of a real variable is different from the usual one given for functions between arbitrary sets, which also requires surjectivity. In this case *injectivity is enough* because we define the inverse on the image of f and not on its whole codomain.

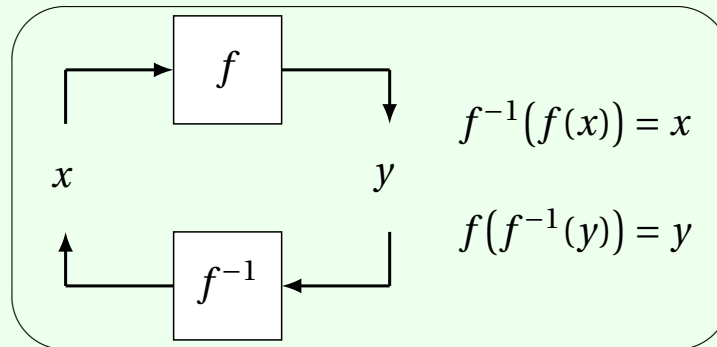
Definition 2: of inverse function

Given an invertible function $f : D \rightarrow \mathbb{R}$, the function that associates to each output $y \in f(D)$ the unique input $x \in D$ such that $f(x) = y$ is called the **inverse function** of f and is denoted by the symbol f^{-1} .

The pair f and f^{-1} is written as:

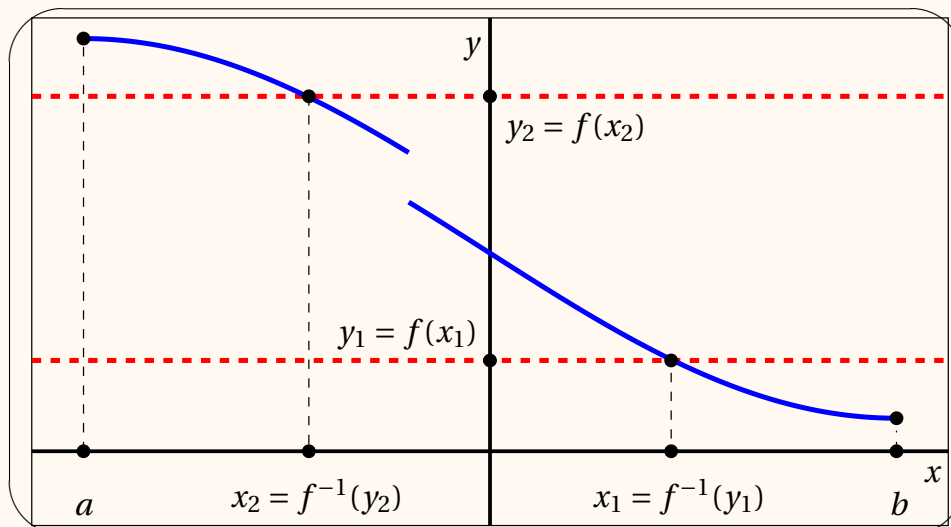
$$f: \begin{cases} y = f(x) \\ x \in D \end{cases} \quad f^{-1}: \begin{cases} x = f^{-1}(y) \\ y \in f(D) \end{cases} \quad (1)$$

The black box of f^{-1} works backwards with respect to that of f , according to the following scheme:



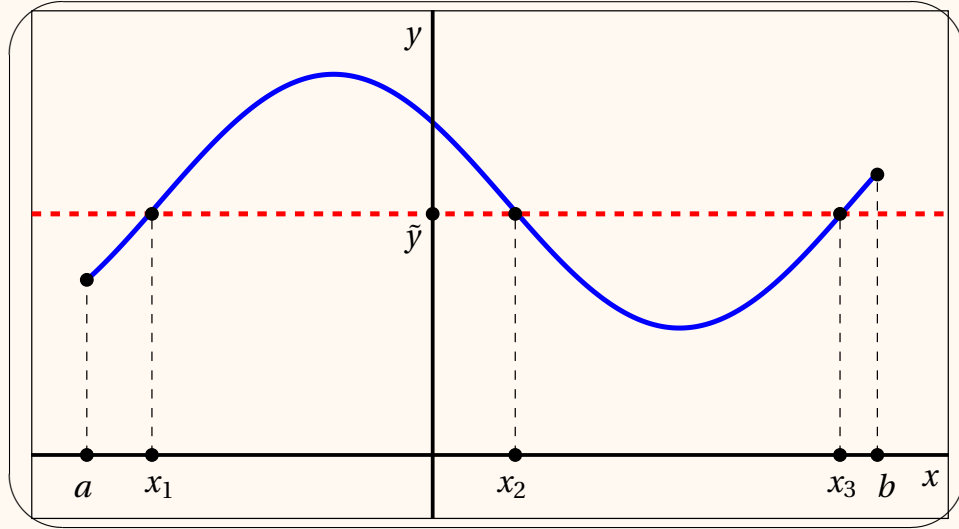
The invertibility condition is equivalent to requiring that the graph of f be intersected in at most one point by every line parallel to the x -axis.

Example 1: graph of an invertible function



Graph of a function that is **invertible** on $[a, b]$, since every line parallel to the x -axis either does not intersect the graph of f or intersects it in exactly one point.

Example 2: graph of a non-invertible function

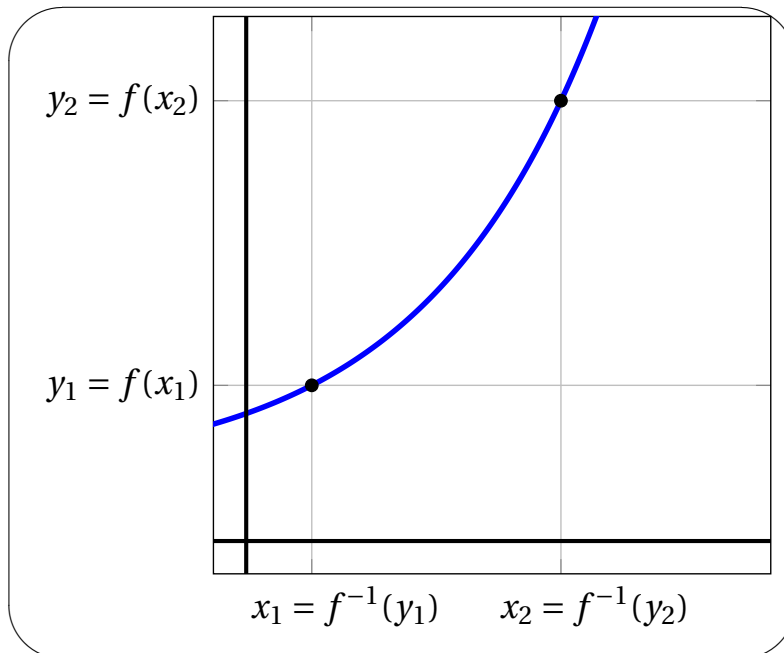


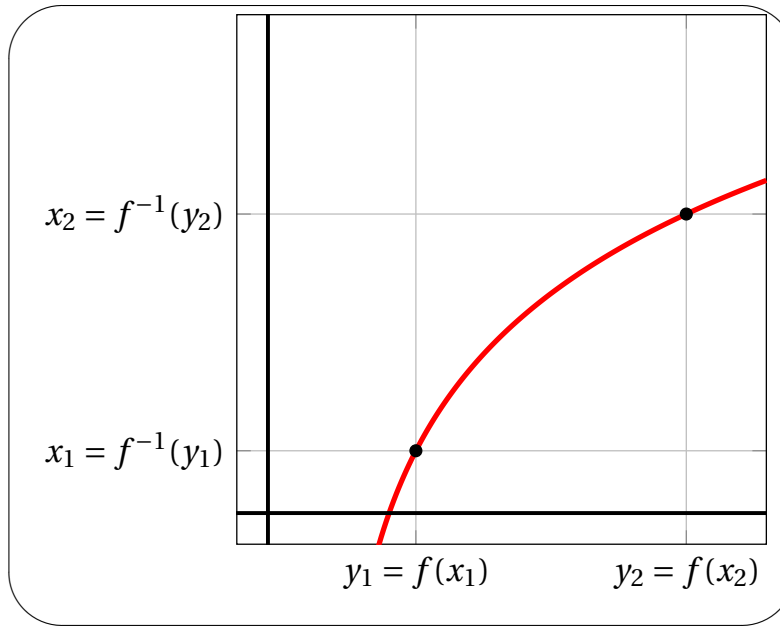
Graph of a function that is **not invertible** on $[a, b]$, since for the indicated value $\tilde{y}$ there are several values x (precisely x_1 , x_2 and x_3) whose image is $\tilde{y}$.

Theorem 1

If a function $f : D \rightarrow \mathbb{R}$ is strictly increasing (decreasing) on D , then it is invertible on D . Moreover, its inverse function is strictly increasing (decreasing).

For the strictly increasing case, graphically we have:





Proof. Consider the case of a function that is strictly increasing on D and two arbitrary values $x_1, x_2 \in D$ (the strictly decreasing case is analogous).

If $x_1 \neq x_2$, then either $x_1 < x_2$ or $x_1 > x_2$. Since f is strictly increasing, we have:

$$\text{if } x_1 < x_2, \text{ then } f(x_1) < f(x_2)$$

$$\text{if } x_1 > x_2, \text{ then } f(x_1) > f(x_2)$$

In both cases $f(x_1) \neq f(x_2)$, therefore f is invertible.

Consider the inverse function f^{-1} , that is, $x = f^{-1}(y)$, and let us prove that it is strictly increasing. Consider two arbitrary values $y_1, y_2 \in f(D)$, with $y_1 < y_2$.

If we had $f^{-1}(y_1) = x_1 \geq x_2 = f^{-1}(y_2)$, since f is strictly increasing we would have $y_1 = f(x_1) \geq f(x_2) = y_2$, that is, $y_1 \geq y_2$. This cannot happen, since it contradicts $y_1 < y_2$ (**contradiction**).

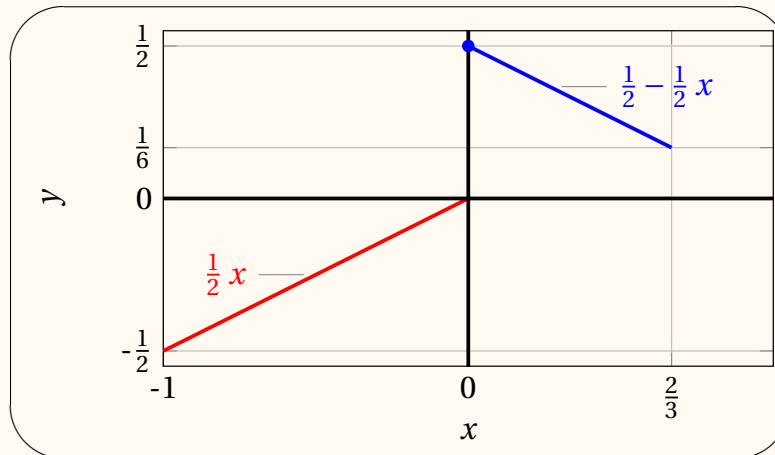
Consequently, we have $f^{-1}(y_1) = x_1 < x_2 = f^{-1}(y_2)$, that is, $f^{-1}(y_1) < f^{-1}(y_2)$, and hence f^{-1} is strictly increasing. □

However, a function can be invertible even without being strictly increasing or decreasing.

Example 3: invertible function that is neither strictly increasing nor strictly decreasing

For example, the following piecewise-defined function is invertible:

$$f(x) = \begin{cases} \frac{1}{2}x & \text{if } -1 \leq x < 0, \\ \frac{1}{2} - \frac{1}{2}x & \text{if } 0 \leq x \leq \frac{2}{3} \end{cases}$$



- Two classes of functions that are certainly not invertible are:

- even (symmetric) functions, since:

$$f(-x) = f(x)$$

- periodic functions, since:

$$f(x + T) = f(x)$$

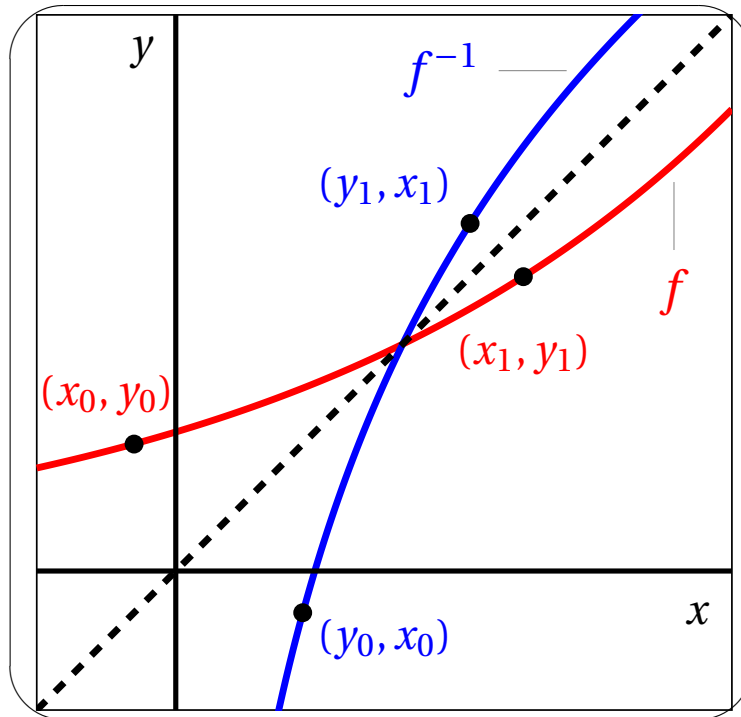
1.1 Graph of the inverse function

- The relations between an invertible function f and its inverse function f^{-1} :

$$f: \begin{cases} y = f(x) \\ x \in D \end{cases} \quad f^{-1}: \begin{cases} x = f^{-1}(y) \\ y \in f(D) \end{cases}$$

indicate that if the point (x_0, y_0) lies on the graph of f , then the point (y_0, x_0) lies on the graph of f^{-1} .

- Since the points (x_0, y_0) and (y_0, x_0) are **symmetric with respect to the bisector** with equation $y = x$, the graph of f^{-1} is obtained from that of f by symmetry with respect to the bisector.



If the analytic expression of f is known and f is invertible, the **analytic expression** of f^{-1} is found by trying to solve for x the equation:

$f(x) = y$ that is, find the input x that produces the output y

Note that it is not always possible to find it, even in cases where the inverse function exists!

Given $a, b \in \mathbb{R}, a \neq 0$, we have:

$$f: \begin{cases} y = ax + b \\ x \in \mathbb{R} \end{cases} \quad f^{-1}: \begin{cases} x = \frac{y-b}{a} \\ y \in \mathbb{R} \end{cases}$$

Example 4: inverse function of an affine function

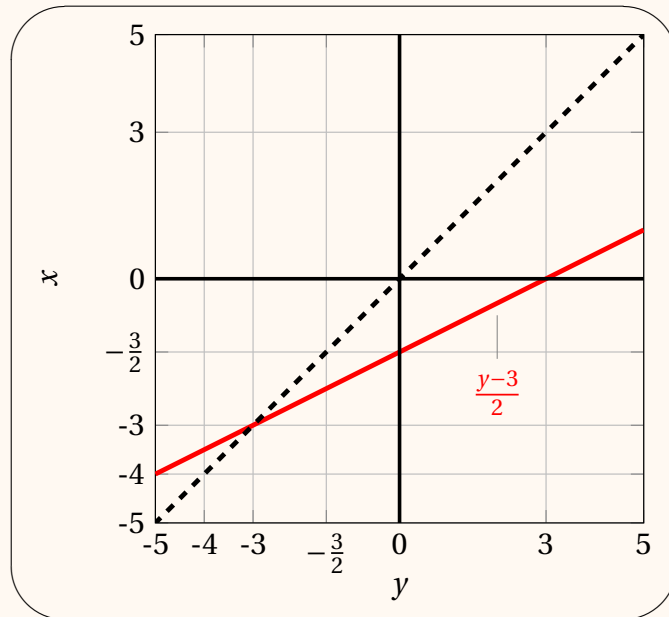
The function $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto 2x + 3$ is strictly increasing, hence invertible on $\mathbb{R}$, and the equation

$$2x + 3 = y \text{ solved for } x, \text{ that is } x = \frac{y-3}{2}$$

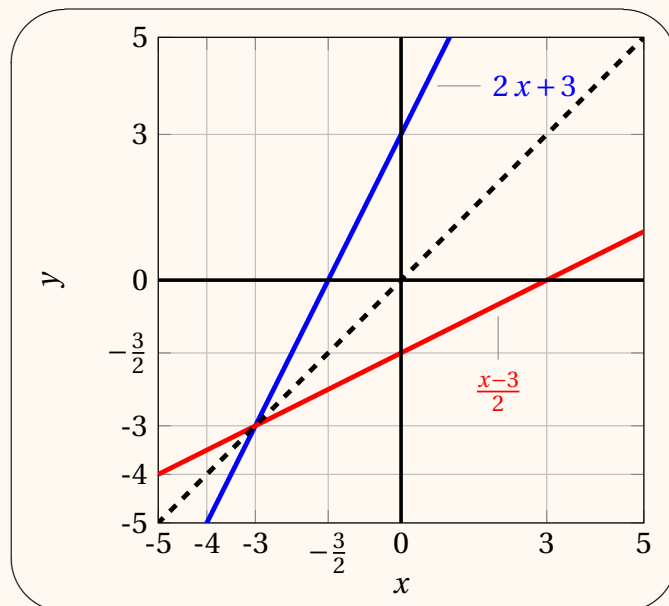
gives for f^{-1} the analytic expression:

$$f^{-1}(y) = \frac{y-3}{2}$$

that is, the function: $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}, y \mapsto \frac{y-3}{2}$.



The graphs $y = 2x + 3$ and $y = \frac{x-3}{2}$ are symmetric with respect to the bisector $y = x$



We have:

$$f: \begin{cases} y = x^2 \\ x \geq 0 \end{cases} \quad f^{-1}: \begin{cases} x = \sqrt{y} \\ y \geq 0 \end{cases}$$

Example 5: inverse function of $f(x) = x^2$

The function $f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$ is strictly increasing on the interval $[0, +\infty)$, hence invertible there; the equation

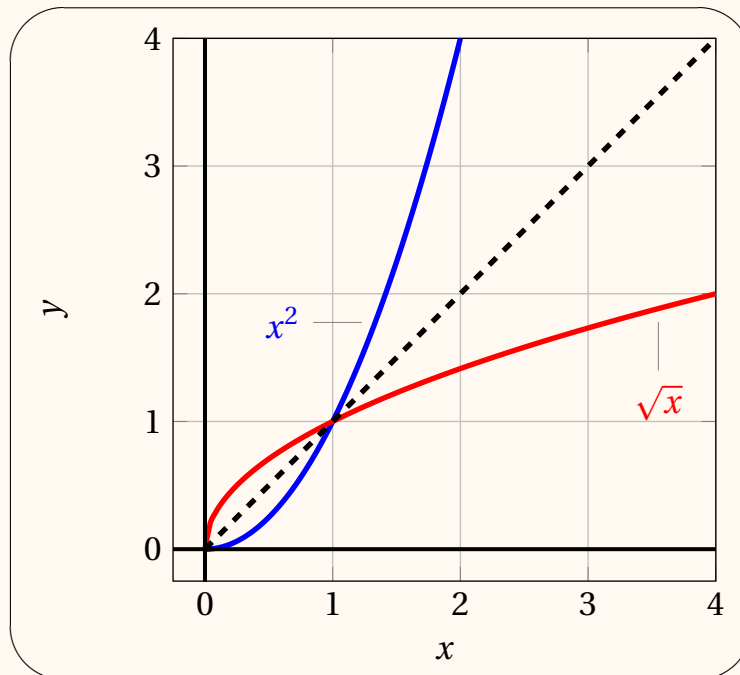
$$x^2 = y \text{ solved for } x, \text{ that is } x = \sqrt{y}$$

gives for f^{-1} the analytic expression

$$f^{-1}(y) = \sqrt{y}$$

that is, the function: $f^{-1}: [0, +\infty) \rightarrow \mathbb{R}, y \mapsto \sqrt{y}$.

The graphs $y = x^2$ and $y = \sqrt{x}$ are symmetric with respect to the bisector $y = x$



Given $a \in \mathbb{R}_+, a \neq 1$, we have:

$$f: \begin{cases} y = a^x \\ x \in \mathbb{R} \end{cases} \quad f^{-1}: \begin{cases} x = \log_a y \\ y > 0 \end{cases}$$

Example 6: inverse function of $f(x) = e^x$

The function $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto e^x$ is strictly increasing on all of $\mathbb{R}$ and hence invertible; the equation

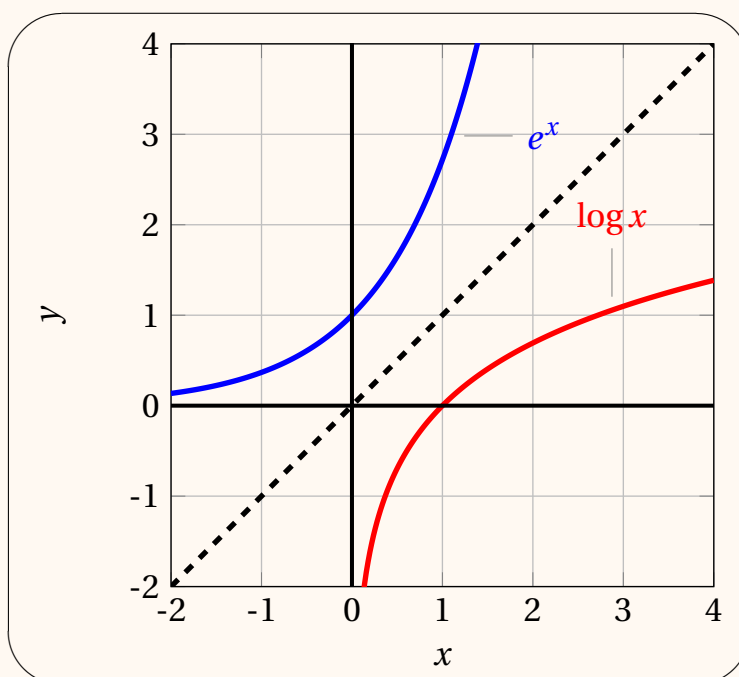
$$e^x = y \text{ solved for } x, \text{ that is } x = \log y$$

gives for f^{-1} the analytic expression

$$f^{-1}(y) = \log y$$

that is, the function: $f^{-1} : (0, +\infty) \rightarrow \mathbb{R}, y \mapsto \log y$.

The graphs $y = e^x$ and $y = \log x$ are symmetric with respect to the bisector $y = x$



- Now consider the function:

$$f(x) = x + e^x$$

Being the sum of two functions that are strictly increasing on all of $\mathbb{R}$, $f(x)$ is strictly increasing and hence invertible on all of $\mathbb{R}$ (see Theorem (??)). However, we would try in vain to solve the equation $x + e^x = y$ for x . In other words, f^{-1} exists, but we cannot write it explicitly.

1.2 Inverse power functions

Given $\alpha \in \mathbb{R}, \alpha \neq 0$, we have:

$$f: \begin{cases} y = x^\alpha \\ x > 0 \end{cases} \quad f^{-1}: \begin{cases} x = y^{\frac{1}{\alpha}} \\ y > 0 \end{cases}$$

If $\alpha > 0$ we have:

$$f: \begin{cases} y = x^\alpha \\ x \geq 0 \end{cases} \quad f^{-1}: \begin{cases} x = y^{\frac{1}{\alpha}} \\ y \geq 0 \end{cases}$$

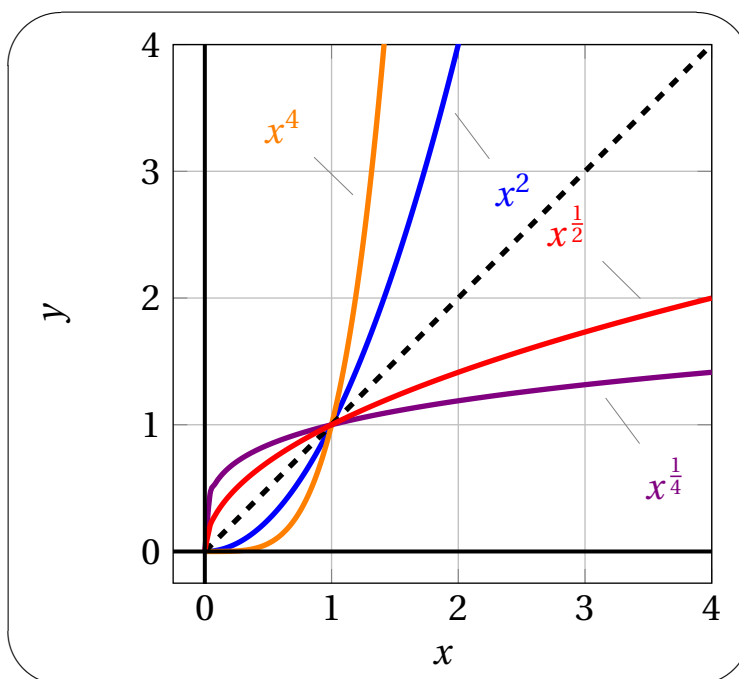
If $\alpha = \frac{m}{n} \in \mathbb{Q}$, with $m \in \mathbb{Z}, n \in \mathbb{N}_+$ odd and coprime, we have:

$$f: \begin{cases} y = x^{\frac{m}{n}} \\ x \in \mathbb{R} \end{cases} \quad f^{-1}: \begin{cases} x = y^{\frac{n}{m}} \\ y \in \mathbb{R} \end{cases}$$

- The **even powers**:

$$x^{2n} \quad \text{with } n = 1, 2, \dots$$

are not invertible on the whole real line, but only on the half-line $x \geq 0$.



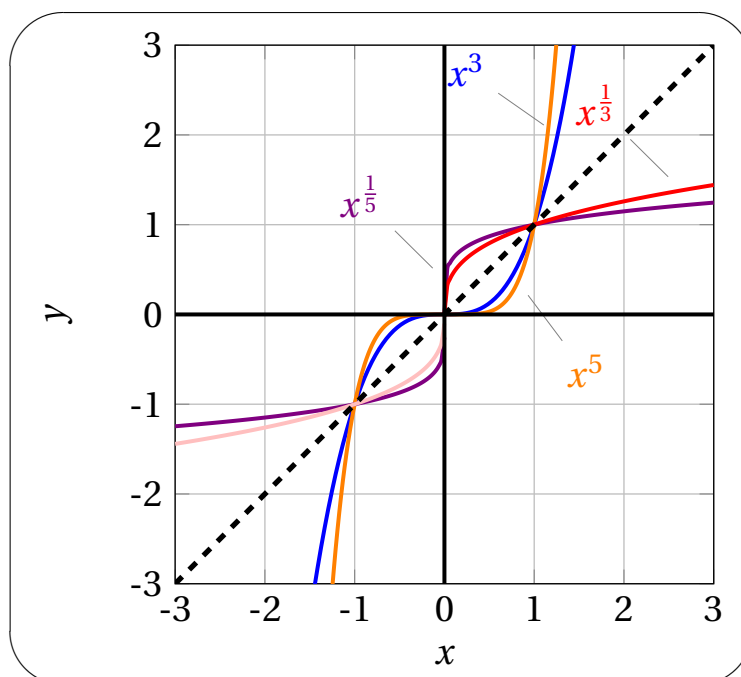
- The **odd powers**:

$$x^{2n+1} \quad \text{with } n = 1, 2, \dots$$

and the **powers with rational exponent**

$$x^{\frac{m}{n}} \quad \text{with } n, m \text{ positive odd integers}$$

being strictly increasing monotone functions, are invertible from $-\infty$ to $+\infty$.



1.3 Inverse trigonometric functions

- Being periodic, the trigonometric functions cannot be invertible. Indeed, for example, the equation

$$y = \sin x$$

has infinitely many solutions if $-1 \leq y \leq 1$ (the output y corresponds to infinitely many inputs), or has no real solutions if $|y| > 1$.

- To speak of inverse functions of sine, cosine and tangent, we will need to restrict ourselves to intervals on which these functions are strictly monotone and therefore invertible.
- An interval on which the sine function is invertible is $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

Definition 3: of arcsine

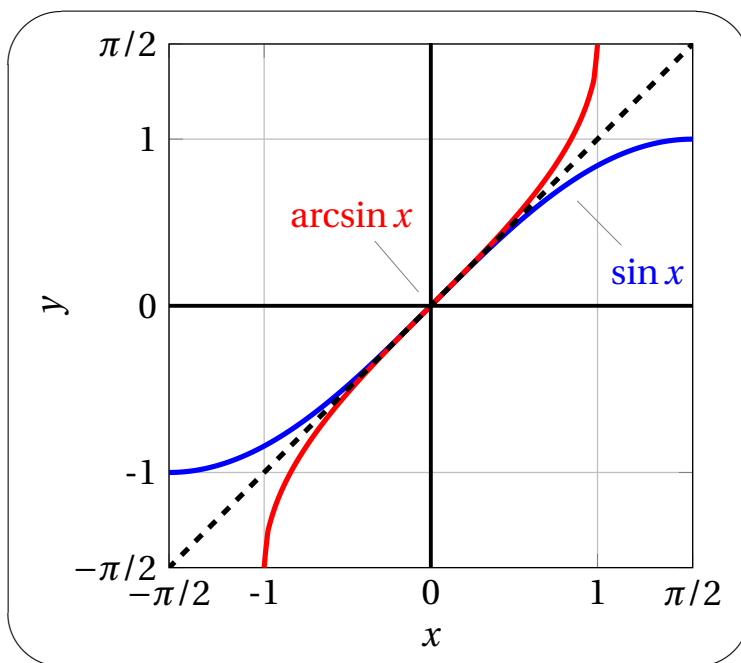
The inverse function of the sine on the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is the **arcsine**:

$$f : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad f : y \mapsto \arcsin y$$

We have:

$$f : \begin{cases} y = \sin x \\ x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \end{cases} \quad f^{-1} : \begin{cases} x = \arcsin y \\ y \in [-1, 1] \end{cases} \quad (2)$$

- The graph of the arcsine is obtained from the arc of the sine curve restricted to the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ (strictly monotone), by symmetry with respect to the bisector $y = x$,



- An interval on which the cosine function is invertible is $[0, \pi]$.

Definition 4: of arccosine

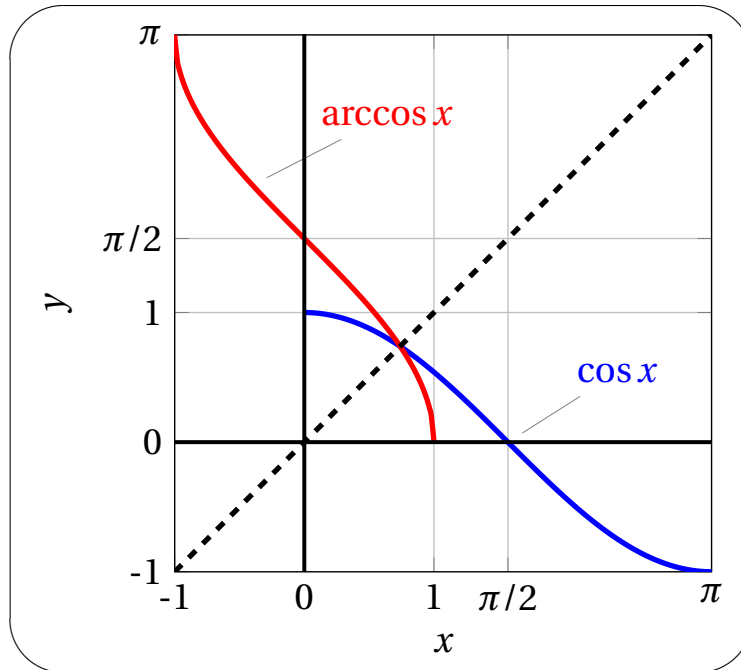
The inverse function of the cosine on the interval $[0, \pi]$ is the **arccosine**:

$$f : [-1, 1] \rightarrow [0, \pi], \quad f : y \mapsto \arccos y$$

We have:

$$f : \begin{cases} y = \cos x \\ x \in [0, \pi] \end{cases} \quad f^{-1} : \begin{cases} x = \arccos y \\ y \in [-1, 1] \end{cases} \quad (3)$$

- The graph of the arccosine is obtained from the arc of the cosine curve restricted to the interval $[0, \pi]$ (strictly monotone), by symmetry with respect to the bisector $y = x$,



- Note that on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ the tangent is strictly monotone and hence invertible. Its inverse function is called the arctangent (arctan), and it is defined on $\mathbb{R}$.

Definition 5: of arctangent

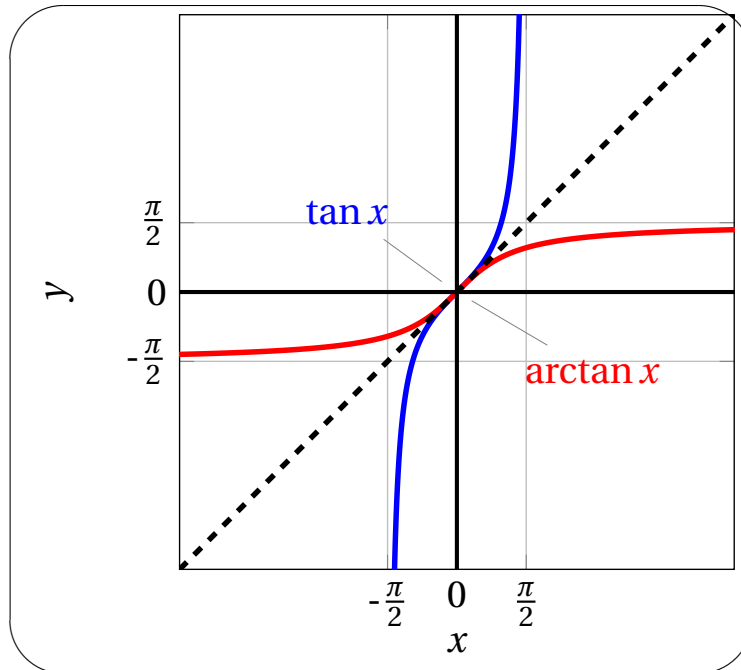
The inverse function of the tangent on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ is the **arctangent**:

$$f: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad f: y \mapsto \arctan y$$

We have:

$$f: \begin{cases} y = \tan x \\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{cases} \quad f^{-1}: \begin{cases} x = \arctan y \\ y \in \mathbb{R} \end{cases} \quad (4)$$

- The graph of the arctangent is obtained from that of the tangent restricted to the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ (strictly monotone), by symmetry with respect to the bisector $y = x$,



- By means of the inverse trigonometric functions, we can express the solutions of a **trigonometric equation or inequality** when it involves angles that are not standard angles.

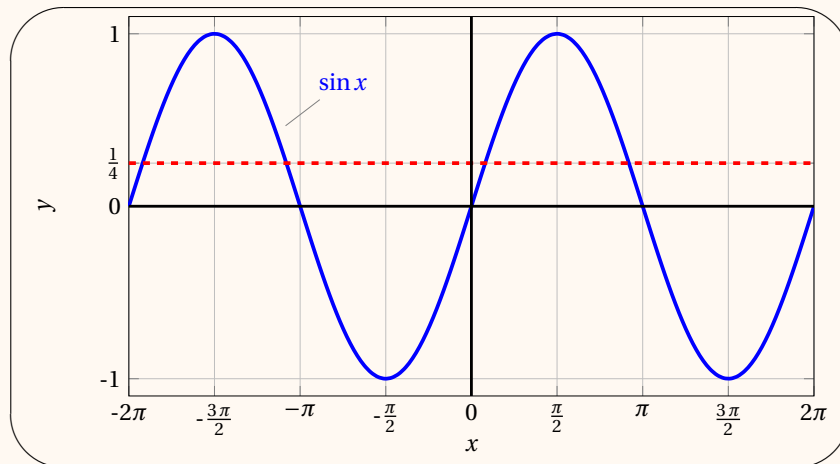
Example 7: Trigonometric equations/inequalities

- The solutions of:

$$\sin x = \frac{1}{4}$$

are:

$$(i) \quad x = \arcsin \frac{1}{4} + 2k\pi; \quad (ii) \quad x = \pi - \arcsin \frac{1}{4} + 2k\pi \quad (k \in \mathbb{Z})$$

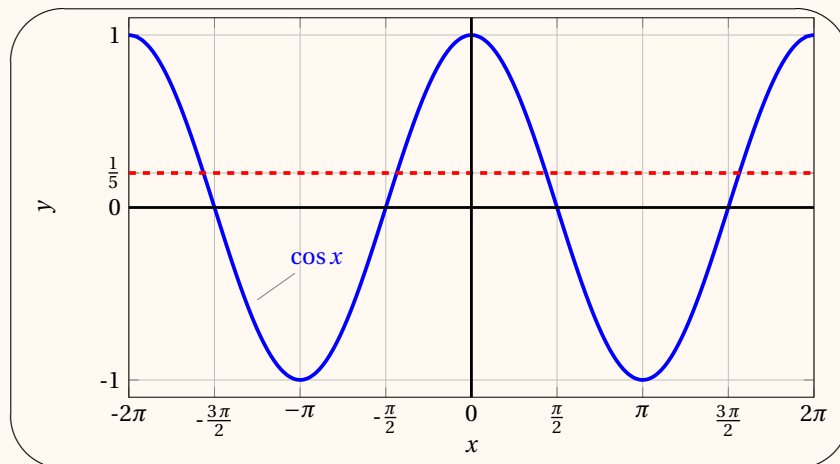


- The solutions of:

$$\cos x < \frac{1}{5}$$

are:

$$\arccos \frac{1}{5} + 2k\pi < x < 2\pi - \arccos \frac{1}{5} + 2k\pi \quad (k \in \mathbb{Z})$$



- The solutions of:

$$\tan x \geq 3$$

are:

$$\arctan 3 + k\pi \leq x < \frac{\pi}{2} + k\pi \quad (k \in \mathbb{Z})$$

1.4 Inverse hyperbolic functions

- Consider the **hyperbolic sine** function:

$$y = \text{Sh } x = \frac{e^x - e^{-x}}{2}$$

It is defined and strictly increasing on all of $\mathbb{R}$, therefore it is invertible.

- To solve the equation for x , multiplying both sides by e^x we obtain:

$$\begin{aligned} 0 &= e^x y - e^x \frac{e^x - e^{-x}}{2} \\ &= 2 e^x y - (e^x e^x - \underbrace{e^x e^{-x}}_{=e^{x-x}=e^0}) \\ &= 2 y e^x - e^{2x} + 1 \\ &= e^{2x} - 2 y e^x - 1 \end{aligned}$$

which is a quadratic equation in the unknown e^x . Setting $t = e^x$ we obtain

$$0 = t^2 - 2 y t - 1$$

We obtain:

$$t = y \pm \sqrt{y^2 + 1} \quad \text{and substituting back} \quad e^x = y \pm \sqrt{y^2 + 1}$$

since $e^x > 0$, the solution with the minus sign must be discarded. Hence we are left with:

$$e^x = \underbrace{y + \sqrt{y^2 + 1}}_{>0, \forall y \in \mathbb{R}} \quad \text{and hence} \quad x = \log \left(y + \sqrt{y^2 + 1} \right)$$

- This is the analytic expression of the inverse function of $\text{Sh } x$, which is called the **inverse hyperbolic sine** (area hyperbolic sine), and is also denoted by $\text{SettSh } y$. It is defined for every real y .

Definition 6: of inverse hyperbolic sine

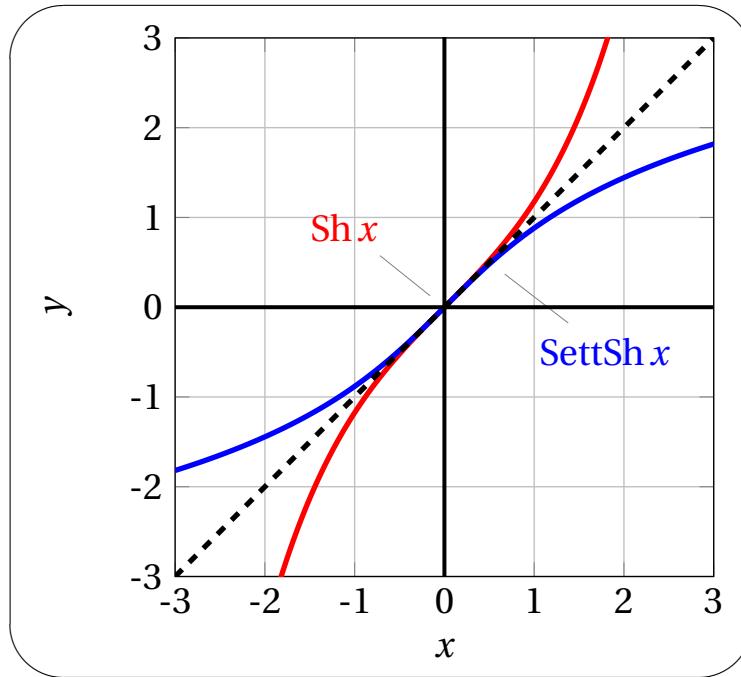
The inverse function of the hyperbolic sine is the **inverse hyperbolic sine**:

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : y \mapsto \text{SettSh } y$$

We have:

$$f : \begin{cases} y = \text{Sh } x \\ x \in \mathbb{R} \end{cases} \quad f^{-1} : \begin{cases} x = \text{SettSh } y \\ y \in \mathbb{R} \end{cases} \quad (5)$$

- The graph of the inverse hyperbolic sine is obtained from that of the hyperbolic sine on $\mathbb{R}$ (strictly monotone), by symmetry with respect to the bisector $y = x$



- Consider the **hyperbolic cosine** function:

$$y = \text{Ch } x = \frac{e^x + e^{-x}}{2}$$

It is defined on all of $\mathbb{R}$, strictly increasing for $x \geq 0$, decreasing for $x \leq 0$. Therefore it is not invertible on all of $\mathbb{R}$.

- Its restriction to $x \geq 0$, however, is invertible. We want to determine the analytic expression of the inverse function of this restriction.
- Proceeding as before, we obtain:

$$e^x = y \pm \sqrt{y^2 - 1}$$

- This time both numbers $y \pm \sqrt{y^2 - 1}$ are positive; recall, however, that we are reasoning only for $x \geq 0$, which is equivalent to choosing the plus sign. Therefore:

$$x = \log\left(y + \sqrt{y^2 - 1}\right)$$

This is the analytic expression of the inverse function of $\text{Ch } x$, which is called the inverse hyperbolic cosine (area hyperbolic cosine), and is also denoted by SettCh . Note that it is defined for $y \geq 1$.

Definition 7: of inverse hyperbolic cosine

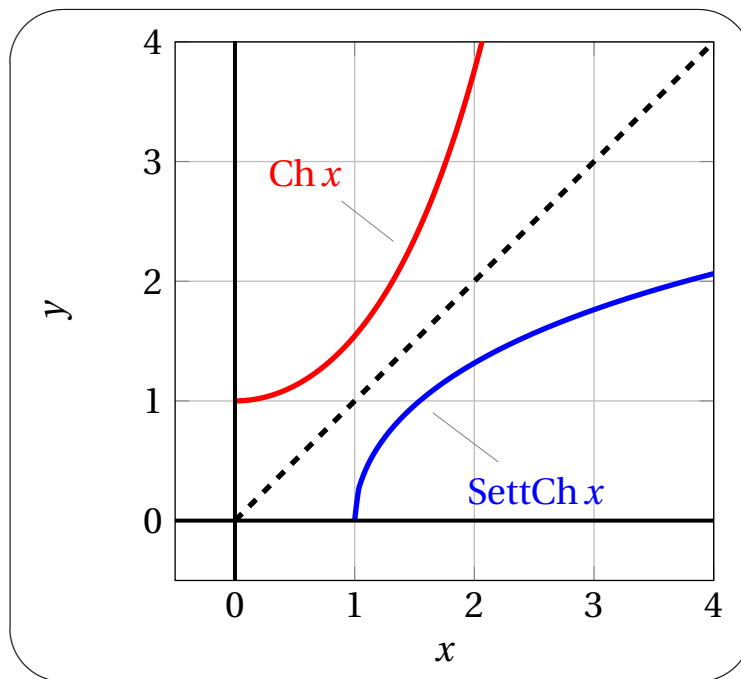
The inverse function of the hyperbolic cosine on the interval $[0, +\infty)$ is the **inverse hyperbolic cosine**:

$$f : [1, +\infty) \rightarrow [0, +\infty), \quad f : y \mapsto \text{SettCh } y$$

We have:

$$f : \begin{cases} y = \text{Ch } x \\ x \in [0, +\infty) \end{cases} \quad f^{-1} : \begin{cases} x = \text{SettCh } y \\ y \in [1, +\infty) \end{cases}$$

- The graph of the inverse hyperbolic cosine is obtained from that of the hyperbolic cosine on $[0, +\infty)$ (strictly monotone), by symmetry with respect to the bisector $y = x$.



Example 8: Hyperbolic equations

- The equation:

$$\operatorname{Sh} x = 2$$

has the unique solution:

$$x = \operatorname{SettSh} 2 = \log(2 + \sqrt{5})$$

- The equation:

$$\operatorname{Ch} x = 3$$

has two solutions:

$$x = \pm \operatorname{SettCh} 3 = \pm \log(3 + 2\sqrt{2})$$

